

Bishwo Biraj Dallakoti

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EDUCATION

Centre College

Danville, KY

B.S. in Computer Science and Data Science

Aug 2024 – May 2028

- **Coursework:** Data Structures, Algorithms, Systems Programming (C), Python, Web Development, Applied Machine Learning, Data Analytics, Data Manipulation, Data Visualization, Calculus II, Discrete Mathematics
- **Award:** Merit-Based Full Tuition Scholarship

SKILLS

Programming Languages: Python, Java, C, JavaScript/TypeScript, SQL, R, HTML, CSS

Frameworks & Web Development: React, Next.js, Node.js, Express, Tailwind CSS, REST APIs, Firebase

Data: PostgreSQL, MySQL, SQLite, Firebase Firestore, Pandas, NumPy, Data Analytics, Data Visualization

Tools: Git, GitHub, Docker, Postman, Excel

EXPERIENCE

Data Science Researcher

June 2025 – August 2025

Centre College

Danville, KY

- Performed applied sports analytics research by analyzing athletic performance of baseball data to generate data-driven insights for competitive decision-making.
- Implemented Principal Component Analysis (PCA) and developed personalized multinomial logistic regression models for individual pitchers, achieving 94.6% pitch-type classification accuracy across five pitch categories using thousands of tracked pitches.
- Collaborated with faculty and peers to transform raw sports performance data into clear, actionable visualizations and analytical reports.

ContractorOops — Full-Stack AI Platform

May 2026 – Present

Software Engineer

Dover, Denver

- Built a modular FastAPI backend supporting the complete contractor workflow, including lead management, quoting, invoicing, scheduling, and client management using domain-driven routers and services.
- Designed an AI agent orchestration layer that converts conversational requests into backend actions such as quote generation, proposal emails, and appointment scheduling using tool registration and contextual memory.
- Integrated third-party services including QuickBooks, Gmail, Google Calendar, SerpAPI, RentCast, and NeetoCal to automate accounting, communications, lead enrichment, and scheduling workflows.
- Implemented secure document ingestion, permission-based retrieval, and document chunking to power AI search and retrieval-augmented generation (RAG) across contractor and client experiences.

PROJECTS

Barber Marketplace — Full-Stack Marketplace & Mobile Platform

Next.js, Expo React Native, TypeScript, Express, PostgreSQL, Stripe, FastAPI, Redis, GPT Image 2

- Architected and launched a two-sided barber marketplace across Next.js web and Expo React Native apps, powered by a strict TypeScript/Express/PostgreSQL monorepo, shared Zod contracts, and 100+ automated tests.
- Engineered end-to-end commerce workflows with concurrency-safe appointment booking, Stripe payments and Connect payouts, subscription tiers, secure authentication, cancellation/refund flows, and operational dashboards for barber scheduling, services, earnings, and customer management.
- Built a location-aware mobile-barber experience with geospatial discovery, travel-time and fee estimation, atomic scheduling buffers, saved addresses, interactive maps, and live GPS journey tracking with continuously recalculated distance and ETA.
- Designed a privacy-first AI Hair Studio using FastAPI, Redis/RQ, MediaPipe, private S3-compatible storage, and GPT Image 2 via fal.ai, enabling consent-managed hairstyle previews, asynchronous generation, cost controls, expiring source images, and direct attachment to bookings.

Dormside Eats — Full-Stack Online Ordering Platform

dormside.xyz Next.js, React, TypeScript, PostgreSQL, Stripe, Nodemailer, Tailwind CSS

- Built and launched a mobile-responsive online ordering platform with Next.js, React, and TypeScript, processing 500+ real customer orders across pickup and delivery.
- Integrated Stripe Payment Intents with server-side payment status, amount, and currency verification, supporting card and cash payments, tips, and delivery fees.
- Developed an authenticated admin dashboard for managing menus, product images, pricing, store availability, and customer orders using signed sessions and middleware authorization.
- Automated customer and administrator notifications through Nodemailer, including dynamically generated branded HTML and PDF receipts for every completed order.